

TransSGPN: Transistor Network Synthesis via Switching Graph Policy Network

Supplementary Material

Anonymous submission

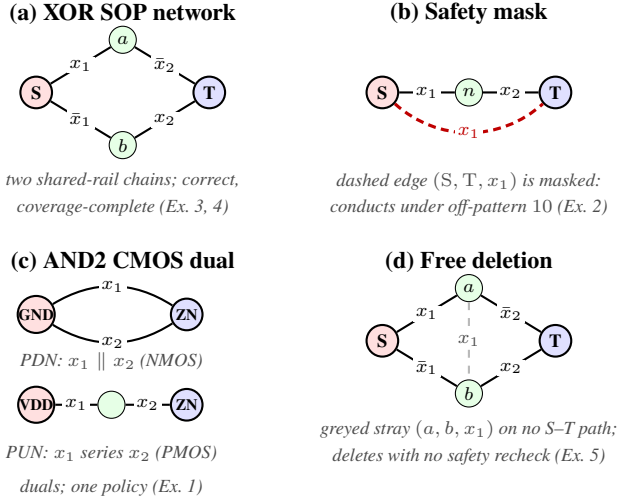


Figure 1: Worked 2-variable circuits for the theoretical results. (a) The XOR SOP network (Example 3): two shared-rail chains, correct and coverage-complete, reachable in any insertion order (Example 4). (b) A safe series chain for x_1x_2 ; the dashed shortcut (S, T, x_1) is rejected by the safety mask because it conducts under the off-pattern 10 (Example 2). (c) The AND2 cell: parallel NMOS pull-down, series PMOS pull-up — duals synthesized by one policy (Example 1). (d) A structurally dead transistor (grey), free to delete with no off-pattern recheck (Example 5). Circles S/T are the source/sink rails; small green nodes are internal.

This supplement collects worked examples, full proofs, and the hyperparameter table referenced in the main paper. Section/Theorem/Example numbers refer to the main paper unless prefixed with “S”.

Worked Examples

Example 1 (Decomposing an AND2 cell). Let $f = x_1x_2$, so $P^+ = \{11\}$ and $P^- = \{00, 01, 10\}$ (Figure 1(c)). The CMOS cell splits into two independent switching problems. The PDN realizes $\neg f$ with on-patterns P^- : it conducts whenever $x_1=0$ or $x_2=0$, the *parallel* pair $x_1 \parallel x_2$ between GND and ZN. The PUN realizes $f(\neg x)$ with on-patterns

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$\{\neg 11\} = \{00\}$: it conducts only when $x_1=0$ and $x_2=0$, the *series* pair between VDD and ZN. The two networks are duals (series $\leftrightarrow$ parallel), synthesized by the *same* policy conditioned on their respective pattern sets.

Example 2 (Monotonicity and incremental safety). Take $f = x_1x_2$, so $P^- = \{00, 01, 10\}$. Suppose the partial network already contains the series pair (Figure 1(b)) $\text{SRC} \xrightarrow{x_1} n \xrightarrow{x_2} \text{SNK}$. This network is *safe*: under 00, 01, 10 at least one of x_1, x_2 is 0, so no $\text{SRC} \rightarrow \text{SNK}$ path conducts. Now add the shortcut $e = (\text{SRC}, \text{SNK}, x_1)$. By monotonicity (Thm. 2) every previously conducting path still conducts, so we test only whether e *by itself* creates an off-pattern path (Cor. 3): under $10 \in P^-$, $x_1 = 1$, so e alone connects SRC to SNK and conducts illegally — e is masked out. Adding $e' = (\text{SRC}, \text{SNK}, x_1x_2)$ would be admissible, since $x_1x_2 = 0$ on all of P^- . We reach this verdict by checking only the new edge against $|P^-| = 3$ patterns, never re-examining the existing chain.

Example 3 (SOP network for a 2-variable function). For $f = x_1\bar{x}_2 + \bar{x}_1x_2$ (XOR), $P^+ = \{10, 01\}$. $G_{\text{SOP}}(f)$ places one 2-transistor chain per on-pattern between the shared rails (Figure 1(a)): chain A is $\text{SRC} \xrightarrow{x_1} a \xrightarrow{\bar{x}_2} \text{SNK}$ and chain B is $\text{SRC} \xrightarrow{\bar{x}_1} b \xrightarrow{x_2} \text{SNK}$. Under 10, chain A conducts and $f = 1$; under 01, chain B conducts. Under 00, 11 each chain has one broken transistor, so nothing conducts. This 4-transistor network is correct but not minimal — XOR admits a 3-transistor bridge — and it serves as the fixed *compensate* network C_f that conditions the state.

Example 4 (The mask never blocks a valid target). Let G^* be the XOR network of Example 3 with edges $\{e_1, e_2\}$ (chain A), $\{e_3, e_4\}$ (chain B), built in the interleaved order e_1, e_3, e_2, e_4 that never completes a chain until the end. At each prefix the partial network is a subgraph of G^* ; by monotonicity any off-pattern path in a prefix would also exist in the safe G^* , so no prefix conducts illegally and the mask rejects none of the four placements — as for all $4! = 24$ orderings. Thus, although path-order supervision presents one linearization, completeness guarantees the policy is free to discover G^* in any order.

Example 5 (A redundant transistor is free to delete). Suppose the policy synthesizes the XOR pull-down of Example 3 (chains A and B) but also emits a stray device $s = (a, b, x_1)$

linking the two internal nodes (Figure 1(d)). Deleting s yields $G' \subseteq G$; by Lemma 6 no off-pattern check is needed, so we re-verify only coverage: chains A and B still carry 10 and 01, hence G' computes the same function. Because s lies on no $\text{SRC} \rightarrow \text{SNK}$ path it is structurally dead, and the reported cost is $T_{\text{eff}} = 4$ rather than 5. On the 6-input NSP benchmark this accounting difference is the gap between a mean of 41.4 and 19.0 transistors.

Proofs

Theorem 1 (Monotonicity of Conducting Paths; Thm. 2). *For any network G and transistor $e \notin E(G)$, $\text{Paths}(G) \subseteq \text{Paths}(G \cup \{e\})$.*

Proof. Adding e can only activate new conducting channels; it cannot deactivate any already-conducting transistor. Hence every path that conducted in G still conducts in $G \cup \{e\}$. $\square$

Corollary 2 (Incremental Safety Checking; Cor. 3). *If G is safe, then $G \cup \{(u, v, \ell)\}$ is safe iff (u, v, ℓ) alone does not create a new $\text{SRC} \rightarrow \text{SNK}$ path under any $\pi \in P^-$.*

Proof. Any P^- -path created in $G \cup \{e\}$ must use e (since G was safe), so checking e alone suffices. $\square$

Theorem 3 (SOP Network Correctness; Thm. 4). *The series-parallel network with one N -transistor series chain per $\pi \in P^+$, all sharing SRC/SNK , correctly implements f .*

Proof. *Coverage:* for $\pi \in P^+$ all literals in the chain evaluate to 1, forming a complete path. *Safety:* for $\pi' \in P^-$ and chain $\pi \in P^+$, since $\pi' \neq \pi$ they differ at some bit j^* , so the j^* -th transistor does not conduct under π' , breaking every path. $\square$

Theorem 4 (Completeness of Safety-Masked Search; Thm. 5). *For any valid G^* implementing f , every permutation of $E(G^*)$ is a valid trajectory under the safety mask.*

Proof. Let $e_1, \dots, e_T$ be any ordering of $E(G^*)$ and $G_t = \{e_1, \dots, e_t\}$. *Base:* $G_0 = \emptyset$ is safe. *Step:* if G_{t-1} is safe, then since $G_t \subseteq G^*$ any P^- -path in G_t would persist in G^* (monotonicity), contradicting G^* 's safety; so G_t is safe and e_t is never rejected. $\square$

Lemma 5 (Deletion preserves safety; Lem. 6). *Let $G' \subseteq G$ by deleting transistors. If G conducts under no off-pattern, then neither does G' .*

Proof. Every path in G' is a path in G ; a violating path in G' would violate G . $\square$

Hyperparameters

Table 1: Architecture and training hyperparameters.

Hyperparameter	Value
MPNN hidden dim d	128
MPNN layers L	3
Head MLP hidden dim	256
Post-MPNN self-attention	yes (4 heads)
Node feature dim d_{node}	4
Edge feature dim d_{edge}	$N + 2$ (5/6/8 for $N = 3/4/5$)
Phase 1 lr (cosine)	$2 \times 10^{-3} \rightarrow 10^{-5}$
Phase 1 batch size	512
Phase 1 epochs	200
Head-1 loss weight	2
Phase 2 lr	10^{-4}
Phase 2 trajectories K	16
PPO clip ε	0.2
Entropy coeff λ_H	0.05
Adaptive-anchor β	0.3
Restarts	2
Max updates per target	200
Max steps T_{max}	20
Phase 3 lr	3×10^{-5}
Phase 3 pairs K	8
ASTRAN SA quality	1
ASTRAN place weights $(w_c, w_m, w_w, w_d, w_g)$	(3, 4, 1, 4, 2)
ASTRAN parallel workers	4